

Experiment 1: Probability Theory

Aim

To calculate the probability of each class (Malignant and Benign) using the Breast Cancer Wisconsin Diagnostic Dataset and predict the class of a new data instance.

Tools Required

- Python
- Pandas
- VS Code

Code

```
import pandas as pd

from sklearn.datasets import load_breast_cancer

data = load_breast_cancer()

X = pd.DataFrame(data.data, columns=data.feature_names)

y = pd.Series(data.target)

class_counts = y.value_counts().sort_index()

total = len(y)

prob_malignant = class_counts[0] / total

prob_benign = class_counts[1] / total

print("Probability of Malignant:", prob_malignant)

print("Probability of Benign:", prob_benign)

if prob_benign > prob_malignant:

    prediction = "Benign"

else:

    prediction = "Malignant"

print("Predicted Class:", prediction)
```

Output

```
E: > brest_cancer.py > ...
E:\ import pandas as pd
2 from sklearn.datasets import load_breast_cancer
3
4 # Load dataset
5 data = load_breast_cancer()
6
7 X = pd.DataFrame(data.data, columns=data.feature_names)
8 y = pd.Series(data.target)
9
10 # Target names
11 print("Classes:", data.target_names)
12
13 # Calculate probabilities
14 class_counts = y.value_counts().sort_index()
15
16 total = len(y)
17
18 prob_malignant = class_counts[0] / total
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Classes: ['malignant' 'benign']

Probability of Malignant: 0.37258347978910367
Probability of Benign: 0.6274165202108963

Predicted class for a new instance: Benign

Result

Probability Theory was successfully implemented using the Breast Cancer Wisconsin Diagnostic Dataset, and the class probabilities were calculated correctly.